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United States Patent	12407536
Kind Code	B2
Date of Patent	September 02, 2025
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System and method for exchanging data between blockchain networks

Abstract

A request is transmitted to a second computing node of a second blockchain network to perform a data interaction between a first computing node of a first blockchain network and the second computing node. Upon receiving an acknowledgement of the request, a first smart contract, an encrypted first network address and a first authentication code is generated. The first smart contract is encrypted using the first authentication code to generate an encrypted first smart contract. The encrypted first smart contract, the encrypted first network address and the first authentication code is transmitted to the second computing node. A digital signature is received indicating an approval of the first smart contract. Thereafter, data objects are transmitted to the second computing node as agreed in the first smart contract.

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Appl. No.:	18/479976
Filed:	October 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250112796 A1	Apr. 03, 2025

Publication Classification

Int. Cl.: H04L9/00 (20220101); H04L9/32 (20060101)

U.S. Cl.:

Field of Classification Search

CPC: H04L (9/50); H04L (9/3226); H04L (9/3247); H04L (9/3239)

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Primary Examiner: Le; Chau

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to network communication, and more specifically to a system and method for exchanging data between blockchain networks.

BACKGROUND

(2) Data interactions in decentralized networks such as a blockchain network are becoming increasingly popular. Data security is of utmost importance in any system that supports online data interactions between computing nodes of the system. However, decentralized networks, by their nature, do not have a central entity that is responsible for administering the security of data interactions within the network. Accordingly, data security and governance is a challenge in decentralized systems such as blockchain networks. These challenges become more pronounced when data is exchanged between two different decentralized networks such as two different blockchain networks.

SUMMARY

(3) The system and method implemented by the system as disclosed in the present disclosure provide technical solutions to the technical problems discussed above by providing secure, technology agnostic, and scalable mechanisms for exchanging data between two different blockchain networks.

(4) For example, the disclosed system and methods provide the practical application of exchanging data (e.g., data objects) between a first blockchain network that supports a first type of data objects and a second blockchain network that supports a second type of data objects. Embodiments of the present disclosure describe an example exchange of data objects between a first computing node of the first blockchain network and a second computing node of the second blockchain network. As described in more detail below, the example exchange of data objects between the first computing node and the second computing includes the first computing node transmitting a first amount of the first type of data objects in exchange for receiving a second amount of the second type of data objects from the second computing node. The first computing node requests the second computing node to perform a data interaction (e.g., a data exchange). In response to receiving an acknowledgement of the request, the first computing node generates and transmits to the second computing node, an encrypted first smart contract, an encrypted first network address and a first authentication code. The first smart contract includes a first transformation rule that specifies how many of the first type of data objects are to be transmitted in exchange for how many of the second type of data objects. The first smart contract is encrypted using the first authentication code and cannot be decrypted without the first authentication code. The first network address is an address within the first blockchain network where the first type of data objects are stored. The first network address is encrypted and cannot be decrypted without an access token. The second computing node decrypts the first smart contract using the first authentication code and reviews the first transformation rule. Upon determining that the first transformation rule is acceptable, the second computing node transmits back a first digital signature along with the first authentication code received from the first computing node. The first computing node receives the first digital signature and the first authentication code from the second computing node and validates the first digital signature by matching the first authentication code received with the first digital signature with the one that was previously transmitted to the second computing node. Upon successfully validating the first digital signature, the first computing node transmits a first access token to the second computing node. The second computing node decrypts the previously received first network address using the first access token and receives the first type of data objects from the first network address within the first blockchain network.

(5) Similarly, the second computing node transmits to the first computing node, an encrypted second network address within the second blockchain network where the second type of data objects are stored, and separately transmits a second access token needed to decrypt the second

network address. The first computing node decrypts the second network address using the second access token and receives the second type of data objects from the second network address within the second blockchain network.

(6) Additionally, transmissions between the first blockchain network and the second blockchain network are encoded using a synthetic DNA synthesis process. This adds another layer of data security.

(7) Thus, the disclosed system and method provide several layers of data security with regard to data exchange between the first blockchain network and the second blockchain network. These data security layers include encrypted smart contracts, encrypted network addresses, validation of digital signatures by matching authentication codes, and encoding data transmissions using a synthetic DNA synthesis process. Thus, the disclosed system and method improve data and network security. In addition, by allowing secure data exchanges between two different blockchain networks, the disclosed system and method save network bandwidth and processing resources which would otherwise be used to detect, identify and resolve data theft. This improves processing and network performance. Further, the disclosed system and method allow data exchange between blockchain networks that support different technologies (e.g., different type of data objects), and thus are technology agnostic. Thus, the disclosed system and methods generally improve blockchain technology and technology related to network security.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 is a schematic diagram of a system, in accordance with certain aspects of the present disclosure; and

(3) FIG. 2 illustrates a flowchart of an example method for exchanging data between a first blockchain network and a second blockchain network, in accordance with one or more embodiments of the present disclosure.

DETAILED DESCRIPTION

(4) FIG. 1 is a schematic diagram of a system **100**, in accordance with certain aspects of the present disclosure. As shown, system **100** includes a first blockchain network **110** and a second blockchain network **150** connected to a network **180**. In one embodiment, the first blockchain network **110** is a part of the network **180** and includes a plurality of computing nodes **112** connected to a portion **180a** of the network **180**. Similarly, the second blockchain network **150** is a part of the network **180** and includes a plurality of computing nodes **152** connected to another portion **180b** of the network **180**. It may be noted that portions **180a** and **180b** of the network **180** may at least partially overlap. In one or more embodiments, while the first blockchain network **110** and the second blockchain network **150** are connected to the same network **180**, the first blockchain network **110** and the second blockchain network **150** operate independently. For example, data interactions performed within the first blockchain network **110** are recorded in the first blockchain network (e.g., in a first blockchain ledger (not shown) associated with the first blockchain network) and not recorded in the second blockchain network **150**. Similarly, data interactions performed within the second blockchain network **150** are recorded in the second blockchain network (e.g., in a second blockchain ledger (not shown) associated with the second blockchain network) and not recorded in the first blockchain network **110**.

(5) In one embodiment, the first blockchain network **110** supports a first type of data objects (e.g., data objects **134**) and the second blockchain network **150** supports a second type of data objects

(e.g., data objects **174**) that is different from the first type of data objects **134**. This means that computing nodes **112** of the first blockchain network **110** may perform data interactions within the first blockchain network **110** based on the first type of data objects **134**. Similarly computing nodes **152** of the second blockchain network **150** may perform data interactions within the second blockchain network **150** based on the second type of data objects **174**. In other words, data interactions occurring within the first blockchain network **110** may be based only on the first type of data objects **134** and data interactions occurring within the second blockchain network **150** may be based only on the second type of data objects **174**. In the context of the present disclosure, the term “data interaction” may refer to transmitting data objects **134**, **174** by a computing node **112**, **152**, receiving data objects **134**, **174** by a computing node **112**, **152**, or exchanging data objects **134**, **174** between two computing nodes **112**, **152**.

(6) In one or more embodiments, the computing nodes **112** and **152** may include, but are not limited to, desktop computers, smartphones, tablet computers, laptop computers, servers and data centers, mainframe computers, virtual reality (VR) headsets, augmented reality (AR) glasses and other hardware devices such as printers, routers, hubs, switches, and memory devices all connected to the network **180**. In one embodiment, at least a portion of the first blockchain network **110** and/or the second blockchain network **150** may be representative of an Information Technology (IT) infrastructure of an organization. In one example, the first blockchain network **110** may be owned and/or managed by a first entity and the second blockchain network **150** may be owned and/or managed by a second entity that is different from and is independent of the first entity.

(7) One or more of the computing nodes **112**, **152** may be operated by users. For example, a computing node **112**, **152** may provide a user interface using which a user may operate the computing node **112**, **152** to perform data interactions within the respective first blockchain network **110** or the second blockchain network **150** to which the computing node **112**, **152** is part of. For example, as shown in FIG. 1, a first user **113** may operate computing node **112a** of the first blockchain network **110** and a second user **153** may operate a computing node **152a** of the second blockchain network **150**.

(8) One or more computing nodes **112**, **152** may be representative of a computing system which hosts software applications that may be installed and run locally or may be used to access software applications running on a server (not shown). The computing system may include mobile computing systems including smart phones, tablet computers, laptop computers, or any other mobile computing devices or systems capable of running software applications and communicating with other devices. The computing system may also include non-mobile computing devices such as desktop computers or other non-mobile computing devices capable of running software applications and communicating with other devices. In certain embodiments, one or more of the computing nodes **112**, **152** may be representative of a server running one or more software applications to implement a respective functionality. In certain embodiments, one or more of the computing nodes **112**, **152** may run a thin client software application where the processing is directed by the thin client but largely performed by a central entity such as a server (not shown).

(9) Network **180**, in general, may be a wide area network (WAN), a personal area network (PAN), a cellular network, or any other technology that allows devices to communicate electronically with other devices. In one or more embodiments, network **180** may be the Internet.

(10) Embodiments of the present describe techniques for exchanging data objects **134**, **174** between computing nodes **112**, **152** of the first blockchain network **110** and the second blockchain network **150**. More specifically, embodiments of the present disclosure describe an example exchange of data objects **134**, **174** between a first computing node **112a** of the first blockchain network **110** and a second computing node **152a** of the second blockchain network **150**. As described in more detail below, the example exchange of data objects **134**, **174** between the first computing node **112a** and the second computing node **152a** includes the first computing node **112a** transmitting a first amount of the first type of data objects **134** belonging to the first user **113** in exchange for receiving

a second amount of the second type of data objects **174** from the second computing node **152a**.

(11) The first computing node **112a** includes a processor **114**, a memory **118**, and a network interface **116**. The first computing node **112a** may be configured as shown in FIG. 1 or in any other suitable configuration.

(12) The processor **114** includes one or more processors operably coupled to the memory **118**. The processor **114** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **114** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **114** is communicatively coupled to and in signal communication with the memory **118**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **114** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **114** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(13) The one or more processors are configured to implement various instructions, such as software instructions. For example, the one or more processors are configured to execute instructions (e.g., first node instructions **136**) to implement the first computing node **112a**. In this way, processor **114** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the first computing node **112a** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The first computing node **112a** is configured to operate as described with reference to FIG. 2. For example, the processor **114** may be configured to perform at least a portion of the method **200** as described in FIG. 2.

(14) The memory **118** comprises a non-transitory computer-readable medium such as one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **118** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(15) The memory **118** is operable to store a first smart contract **122**, a first encrypted network address **124**, a first authentication code **126**, a first data file **132**, a first type of data objects **134** and first node instructions **136**. The first node instructions **136** may include any suitable set of instructions, logic, rules, or code operable to execute the first computing node **112a**.

(16) The network interface **116** is configured to enable wired and/or wireless communications. The network interface **116** is configured to communicate data between the first computing node **112a** and other devices, systems, or domains (e.g., second computing node **152a** and/or other computing nodes **112** and **152**). For example, the network interface **116** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **114** is configured to send and receive data using the network interface **116**. The network interface **116** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(17) The second computing node **152a** includes a processor **154**, a memory **158**, and a network interface **156**. The second computing node **152a** may be configured as shown in FIG. 1 or in any other suitable configuration.

(18) The processor **154** includes one or more processors operably coupled to the memory **158**. The processor **154** is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-

programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **154** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The processor **154** is communicatively coupled to and in signal communication with the memory **158**. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **154** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **154** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components.

(19) The one or more processors are configured to implement various instructions, such as software instructions. For example, the one or more processors are configured to execute instructions (e.g., second node instructions **176**) to implement the second computing node **152a**. In this way, processor **154** may be a special-purpose computer designed to implement the functions disclosed herein. In one or more embodiments, the second computing node **152a** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The second computing node **152a** is configured to operate as described with reference to FIG. 2. For example, the processor **154** may be configured to perform at least a portion of the method **200** as described in FIG. 2.

(20) The memory **158** comprises a non-transitory computer-readable medium such as one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **158** may be volatile or non-volatile and may comprise a read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM).

(21) The memory **158** is operable to store a second smart contract **162**, a second encrypted network address **164**, a second authentication code **166**, a second data file **172**, a second type of data objects **174** and second node instructions **176**. The second node instructions **176** may include any suitable set of instructions, logic, rules, or code operable to execute the second computing node **152a**.

(22) The network interface **156** is configured to enable wired and/or wireless communications. The network interface **156** is configured to communicate data between the second computing node **152a** and other devices, systems, or domains (e.g., first computing node **112a** and/or other computing nodes **112** and **152**). For example, the network interface **156** may comprise a Wi-Fi interface, a LAN interface, a WAN interface, a modem, a switch, or a router. The processor **154** is configured to send and receive data using the network interface **156**. The network interface **156** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(23) It may be noted that each of the computing nodes **112** and **152** may be implemented like the first computing node **112a** and the second computing node **152a** shown in FIG. 1. For example, each of the computing nodes **112** and **152** may have a respective processor and a memory that stores data and instructions to perform operations discussed above.

(24) The first computing node **112a** of the first blockchain network **110** may be configured to perform a data interaction with the second computing node **152a** of the second blockchain network **150** to exchange data objects with the second computing node **152a**. This exchange of data objects may include the first computing node **112a** transmitting a first amount of the first type of data objects **134** in exchange for receiving a second amount of the second type of data objects **174** from the second computing node **152a**. In this context, the first user **113** may store a first data file **132** at the first computing node **112a** owned and/or operated by the first user **113**. In the first data file **132**, the first user **113** may store the first type of data objects **134** owned by the first user **113**. Similarly, the second user **153** may store a second data file **172** at the second computing node **152a** owned

and/or operated by the second user **153**. In the second data file **172**, the second user **153** may store the second type of data objects **174** owned by the second user **153**.

(25) To initiate the data exchange with the second computing node **152a**, the first computing node **112a** may be configured to transmit to the second computing node **152a**, a request **119** to perform a data interaction with the second computing node **152**. The request **119** may indicate that the requested data interaction relates to exchanging data (e.g., data objects) between the first user **113** and the second user **153**. In one embodiment, the first user **113** may trigger the generation and transmission of the request **119** by the first computing node **112a**. In response to receiving the request **119** from the first computing node **112a**, the second computing node **152a** may generate and transmit an acknowledgement **159** to the first computing node **112a** indicating that the second computing node **152a** is ready to perform the requested data interaction with the first computing node **112a**. In one embodiment, the acknowledgement may be automatically generated by the second computing node **152a** or manually triggered by the second user **153**. For example, the second user **153** review the request **119** received by the second computing node **152a** and trigger the generation and transmission of the acknowledgement **159** to the first computing node **112a**.

(26) Upon receiving the acknowledgement **159** from the second computing node **152a**, the first computing node **112a** may be configured to generate a first smart contract **122** that includes information relating to the data interaction. For example, the first smart contract **122** may include a first transformation rule **123** indicating that the first computing node **112a** is to transmit a first amount of the first type of data objects belonging to the first user **113** in exchange for a second amount of the second type of data objects belonging to the second user **153** that is to be received from the second computing node **152a**. In addition, the first computing node **112a** generates a first authentication code **126** that is unique to the data interaction initiated between the first computing node **112a** and the second computing node **152a**. The first computing node **112a** encrypts the first smart contract **122** using the first authentication code **126**. In one embodiment, the first computing node **112a** encrypts the first smart contract **122** based on an encrypt-decrypt algorithm previously agreed with the second computing node **152a**. As further described below, this allows the second computing node **152a** to decrypt the first smart contract **122** using the first authentication code **126** based on the same encrypt-decrypt algorithm previously agreed with the first computing node **112a**. Essentially, the first authentication code **126** acts as a unique key that can be used by the second computing node **152a** to access and review the first smart contract **122**. The first authentication code **126** may include any known code such as an alphanumeric code. In addition, the first computing node **112a** generates an encrypted first network address **124** of a memory location within the first blockchain network **110** where the first type of data objects **134** associated with the first user **113** are stored. The first type of data objects **134** associated with the first user **113** may be stored at the first computing node **112a** or at any other computing node **112** of the first blockchain network **110**.

(27) The first computing node **112a** may transmit the first smart contract **122** encrypted using the first authentication code **126**, the encrypted first network address **124**, and the first authentication code **126** to the second computing node **152a**. In one embodiment, the first computing node **112a** transmits the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** in a single encoded first data file **120**. In alternative embodiments, the first computing node **112a** transmits the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** as separate data files.

(28) Upon receiving the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** from the first computing node **112a**, the second computing node **152a** may be configured to decrypt the first smart contract **122** using the first authentication code **126**. As described above, the second computing node **152a** may decrypt the first smart contract **122** based on the encrypt-decrypt algorithm previously agreed with the first computing node **112a**. After decrypting the first smart contract **122**, the second computing node **152a** may

extract the first transformation rule **123** included in the first smart contract **122** and determine whether the first transformation rule **123** is acceptable. In one embodiment, the second computing node **152a** may determine whether the first transformation rule **123** is acceptable based on a pre-configured rule set for the second blockchain network **150** relating to data exchanges between a computing node **152** of the second blockchain network **150** and a computing node **112** of the first blockchain network. For example, the pre-configured rule set for the second blockchain network **150** may specify that at least X amount of the first type of data objects is to be received for a unit amount of the second type of data objects. The second computing node **152a** analyzes the first transformation rule **123** based on this pre-configured rule to determine whether the first transformation rule **123** is acceptable. As mentioned above, the first transformation rule **123** indicates that the first computing node **112a** is to transmit a first amount of the first type of data objects belonging to the first user **113** in exchange for a second amount of the second type of data objects **174** belonging to the second user **153** that is to be received from the second computing node **152a**. When the first transformation rule **123** satisfies the pre-configured rule set for the second blockchain network **150**, the second computing node **152a** determines that the first transformation rule **123** is acceptable. For example, based on the first transformation rule **123**, when at least X amount of the first type of data objects **134** is to be received for a unit amount of the second type of data objects **174**, the second computing node **152a** determines that the first transformation rule **123** is acceptable and approves the first transformation rule **123**.

(29) It may be noted that the encrypted first network address **124** cannot be decrypted by the second computing node **152a** at this stage. The encrypted first network address **124** is configured to be decrypted by a unique access token (e.g., first access token **130**) which, as described below, the first computing node **112a** is configured to transmit to the second computing node **152a** at a later stage.

(30) In response to determining that the first transformation rule **123** is acceptable, the second computing node generates and transmits a first digital signature **168** to the first computing node **112a**. The first digital signature **168** indicates to the first computing node **112a** that the second computing node **152a** has approved the first transformation rule **123** that was included in the first smart contract **122**. In one embodiment, the second computing node **152a** transmits the first digital signature **168** along with the first authentication code **126** that was previously received by the second computing node **152a** from the first computing node **112a**. This allows the first computing node **112a** to validate the first digital signature **168** and map it to the particular data interaction that was requested via request **119** (e.g., since the first authentication code **126** is unique to the particular data interaction). For example, upon receiving the first digital signature **168** and the first authentication code **126**, the first computing node **112a** compares the first authentication code **126** received from the second computing node **152a** to the first authentication code **126** that was generated and transmitted by the first computing node **112a**. When the two codes match, the first computing node **112a** determines that the first digital signature **168** has been validated and that the first digital signature **168** is an approval of the first transformation rule **123** included in the first smart contract **122**. In an additional or alternative embodiment, the second computing node **152a** transmits the first digital signature **168** and the first authentication code **126** as part of the first smart contract **122** that was received from the first computing node **112a**. In other words, to indicate an approval of the first transformation rule **123**, the second computing node **152a** transmits back the first smart contract **122** along with the first digital signature **168** and the first authentication code **126**.

(31) After validating the first digital signature **168**, the first computing node transmits a first access token **130** to the second computing node **152a**. The first access token **130** is a key that can be used to decrypt the encrypted first network address **124** that was previously transmitted to the second computing node **152a**. As described above, the encrypted first network address **124** is of a memory location within the first blockchain network **110** where the first type of data objects **134** associated

with the first user **113** are stored. The second computing node **152a** receives the first access token **130** and decrypts the encrypted first network address **124** using the first access token **130** to determine the first network address **124**. Thereafter, the second computing node **152a** requests the first computing node **112a** to transmit the first amount of the first type of data objects **134** belonging to the first user **113** from the first network address **124**. For example, the second computing node **152a** transmits a request for the first amount of the first type of data objects **134** along with the first network address **124**. In response to receiving the request for the first type of data objects **134**, the first computing node **112a** initiates a transmission of the first amount of the first type of data objects **134** to the second computing node **152a**. In one embodiment, the second computing node **152a** transmits the request for the first type of data objects **134** to an exchange computing node (an exchange server in the second blockchain network **150**) in the second blockchain network **150** that is configured to coordinate data exchange between the first blockchain network **110** and the second blockchain network **150**. In an alternative or additional embodiment, the second computing node **152a** performs operations relating to the exchange computing node.

(32) The second computing node **152a** may be configured to transmit a second network address **164** of a memory location within the second blockchain network **150** where the second data objects **174** owned by the second user **153** are stored. In one embodiment, the second computing node **152a** transmits an encrypted second network address **164** and separately transmits a second access token **170** to the first computing node **112a**. The second access token **170** is a key that can be used to decrypt the encrypted second network address **164** that was previously transmitted to the first computing node **112a**. The first computing node **112a** receives the second access token **170** and decrypts the encrypted second network address **164** using the second access token **170** to determine the second network address **164**. Thereafter, the first computing node **112a** requests the second computing node **152a** to transmit the second amount of the second type of data objects **174** belonging to the second user **153** from the second network address **164**. For example, the first computing node **112a** transmits a request for the second amount of the second type of data objects **174** along with the second network address **164**. In response to receiving the request for the second type of data objects **174**, the second computing node **152a** initiates a transmission of the second amount of the second type of data objects **174** to the first computing node **112a**. In one embodiment, the first computing node **112a** transmits the request for the second type of data objects **174** to an exchange computing node (e.g., exchange server) within the first blockchain network **110** that is configured to coordinate data exchange between the first blockchain network **110** and the second blockchain network **150**. In an alternative or additional embodiment, the first computing node **112a** performs operations relating to the exchange computing node of the first blockchain network **110**. In one embodiment, the second computing node **152a** transmits the encrypted second network address **164** and the second access token **170** after approving the first transformation rule **123** included in the first smart contract **122**.

(33) In one or more embodiments, the first computing node **112a** and the second computing node **152a** may engage in an electronic negotiation process to determine how many of the first type of data objects **134** belonging to the first user **113** are to be exchanged for how many of the second type of data objects **174** belonging to the second user **153**. For example, after transmitting the acknowledgement **159** to the first computing node **112a** or after receiving the first smart contract **122** from the first computing node **112a**, the second computing node **152a** generates a second smart contract **162** that includes information relating to the data interaction requested by the first computing node **112a**. For example, the second smart contract **162** may include a second transformation rule **163** indicating that the second computing node **152a** is to transmit a third amount of the second type of data objects **174** belonging to the second user **153** in exchange for the first amount of the first type of data objects **134** belonging to the first user **113** that is to be received from the first computing node **112a**. In one embodiment, the third amount of the second type of data objects **174** associated with the second transformation rule **163** may be same as the second

amount of the second type of data objects **174** associated with the first transformation rule **123**. In other words, the second transformation rule **163** is same as the first transformation rule **123**.

(34) In an alternative embodiment, the third amount of the second type of data objects **174** associated with the second transformation rule **163** may be different from the second amount of the second type of data objects **174** associated with the first transformation rule **123**. For example, the second computing node **152a** may determine that the first transformation rule **123** included in the first smart contract **122** is not agreeable, and in response, generates the second smart contract **162** that includes the second transformation rule **163** different from the first transformation rule **123**.

(35) In addition to generating the second smart contract **162**, the second computing node **152a** generates a second authentication code **166** that is unique to the data interaction initiated between the first computing node **112a** and the second computing node **152a**. The second computing node **152a** encrypts the second smart contract **162** using the second authentication code **166**. In one embodiment, the second computing node **152a** encrypts the second smart contract **162** based on an encrypt-decrypt algorithm previously agreed with the first computing node **112a**. As further described below, this allows the first computing node **112a** to decrypt the second smart contract **162** using the second authentication code **166** based on the same encrypt-decrypt algorithm previously agreed with the second computing node **152a**. Essentially, the second authentication code **166** acts as a unique key that can be used by the first computing node **112a** to access and review the second smart contract **162**. The second authentication code **166** may include any known code such as an alphanumeric code. In addition, the second computing node **152a** generates an encrypted first network address **164** of a memory location within the second blockchain network **150** where the second type of data objects **174** associated with the second user **153** are stored. The second type of data objects **174** associated with the second user **153** may be stored at the second computing node **152a** or at any other computing node **152** of the second blockchain network **150**.

(36) The second computing node **152a** may transmit the second smart contract **162** encrypted using the second authentication code **166**, the encrypted second network address **164**, and the second authentication code **166** to the first computing node **112a**. In one embodiment, the second computing node **152a** transmits the encrypted second smart contract **162**, the encrypted second network address **164**, and the second authentication code **166** in a single encoded second data file **160**. In alternative embodiments, the second computing node **152a** transmits the encrypted second smart contract **162**, the encrypted second network address **164**, and the second authentication code **166** as separate data files.

(37) Upon receiving the encrypted second smart contract **162**, the encrypted second network address **164**, and the second authentication code **166** from the second computing node **152a**, the first computing node **112a** may be configured to decrypt the second smart contract **162** using the second authentication code **166**. As described above, the first computing node **112a** may decrypt the second smart contract **162** based on the encrypt-decrypt algorithm previously agreed with the second computing node **152a**. After decrypting the second smart contract **162**, the first computing node **112a** may extract the second transformation rule **173** included in the second smart contract **162** and determine whether the second transformation rule **173** is acceptable. In one embodiment, when the second transformation rule **173** is same as the first transformation rule **123** that was previously transmitted by the first computing node **112a** as part of the first smart contract **122**, the first computing node determines that the second transformation rule **163** is acceptable and approves the second smart contract **162**.

(38) In an alternative embodiment, when the second transformation rule **163** is different from the first transformation rule **123**, the first computing node **112a** may determine whether the second transformation rule **173** is acceptable based on a pre-configured rule set for the first blockchain network **110** relating to data exchanges between a computing node **112** of the first blockchain network **110** and a computing node **152** of the second blockchain network **150**. For example, the pre-configured rule set for the first blockchain network **110** may specify that at least Y amount of

the second type of data objects **174** is to be received for a unit amount of the first type of data objects **134**. The first computing node **112a** analyzes the second transformation rule **173** based on this pre-configured rule to determine whether the second transformation rule **173** is acceptable. As mentioned above, the first transformation rule **173** indicates that the second computing node **152a** is to transmit a third amount of the second type of data objects **174** belonging to the second user **153** in exchange for a first amount of the first type of data objects **134** belonging to the first user **113** that is to be received from the first computing node **112a**. When the second transformation rule **173** satisfies the pre-configured rule set for the first blockchain network **110**, the first computing node **112a** determines that the second transformation rule **173** is acceptable. For example, based on the second transformation rule **173**, when at least Y amount of the second type of data objects **174** is to be received for a unit amount of the first type of data objects **134**, the first computing node **112a** determines that the second transformation rule **173** is acceptable and approves the second transformation rule **173**.

(39) In certain embodiments, the first computing node **112a** and the second computing node **152a** may exchange two or more smart contracts **122**, **162** before both the first computing node **112a** and the second computing node **152a** agree on a particular transformation rule **123**, **163**.

(40) In response to determining that the second transformation rule **163** is acceptable, the first computing node generates and transmits a second digital signature **128** to the second computing node **152a**. The second digital signature **128** indicates to the second computing node **152a** that the first computing node **112a** has approved the second transformation rule **163** that was included in the second smart contract **162**. In one embodiment, the first computing node **112a** transmits the second digital signature **128** along with the second authentication code **166** that was previously received by the first computing node **112a** from the second computing node **152a**. This allows the second computing node **152a** to validate the second digital signature **128** and map it to the particular data interaction that was requested via request **119** (e.g., since the second authentication code **166** is unique to the particular data interaction). For example, upon receiving the second digital signature **128** and the second authentication code **166**, the second computing node **152a** compares the second authentication code **166** received from the first computing node **112a** to the second authentication code **166** that was generated and transmitted by the second computing node **152a**. When the two codes match, the second computing node **152a** determines that the second digital signature **128** has been validated and that the second digital signature **128** is an approval of the second transformation rule **163** included in the second smart contract **162**. In an additional or alternative embodiment, the first computing node **112a** transmits the second digital signature **128** and the first authentication code **126** as part of the second smart contract **162** that was received from the second computing node **152a**. In other words, to indicate an approval of the second transformation rule **163**, the first computing node **112a** transmits back the second smart contract **162** along with the second digital signature **128** and the second authentication code **166**.

(41) After validating the second digital signature **128**, the second computing node **152a** transmits a second access token **170** to the first computing node **112a**. The second access token **170** is a key that can be used to decrypt the encrypted second network address **164** that was previously transmitted to the first computing node **112a**. As described above, the encrypted second network address **164** is of a memory location within the second blockchain network **150** where the second type of data objects **174** associated with the second user **153** are stored. The first computing node **112a** receives the second access token **170** and decrypts the encrypted second network address **164** using the second access token **170** to determine the second network address **164**. Thereafter, the first computing node **112a** requests the second computing node **152a** to transmit the third amount (or second amount whichever is the case) of the second type of data objects **174** belonging to the second user **153** from the second network address **164**. For example, the first computing node **112a** transmits a request for the third amount (or second amount) of the second type of data objects **174** along with the second network address **164**. In response to receiving the request for the second type

of data objects **174**, the second computing node **152a** initiates a transmission of the third amount (or second amount) of the second type of data objects **174** to the first computing node **112a**. In one embodiment, the first computing node **112a** transmits the request for the second type of data objects **174** to an exchange node (not shown) in the first blockchain network **110** that is configured to coordinate data exchange between the first blockchain network **110** and the second blockchain network **150**. In an alternative or additional embodiment, the first computing node **112a** performs operations relating to the exchange computing node.

(42) In one or more embodiments, the transmissions between the first computing node **112a** and the second computing node **152a** are encoded using synthetic (e.g., artificial) DNA synthesis **175** process. For example, the first computing node **112a** encodes the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** using synthetic DNA synthesis to generate an encoded first data file **120**, wherein the encoded first data file **120** includes one or more synthetic DNA sequences that represent encoded data. The first computing node **112a** transmits the single encoded first data file **120** to the second computing node **152a**. The second computing node **152a** decodes the one or more synthetic DNA sequences in the encoded first data file **120** to extract the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126**.

(43) Similarly, the second computing node **152a** encodes the encrypted second smart contract **162**, the encrypted second network address **164**, and the second authentication code **166** to generate the single encoded second data file **160**, wherein the encoded second data file **160** includes one or more synthetic DNA sequences that represent the encoded data. The second computing node **152a** transmits the encoded second data file **160** to the first computing node **112a**. The first computing node **112a** decodes the one or more synthetic DNA sequences in the encoded second data file **160** to extract the encrypted second smart contract **162**, the encrypted second network address **164**, and the second authentication code **166**.

(44) FIG. 2 illustrates a flowchart of an example method **200** for exchanging data between a first blockchain network **110** and a second blockchain network **150**, in accordance with one or more embodiments of the present disclosure. Method **200** may be performed by the first computing node **112a** of the first blockchain network **110** shown in FIG. 1.

(45) At operation **202**, first computing node **112a** transmits, to the second computing node **152a** of the second blockchain network **150**, a request **119** to perform a data interaction with the second computing node **152a**, wherein the data interaction comprises exchange of data (e.g., data objects **134**, **174**) between the first computing node **112a** and the second computing node **152a**.

(46) As described above, the first computing node **112a** of the first blockchain network **110** may be configured to perform a data interaction with the second computing node **152a** of the second blockchain network **150** to exchange data objects with the second computing node **152a**. This exchange of data objects may include the first computing node **112a** transmitting a first amount of the first type of data objects **134** in exchange for receiving a second amount of the second type of data objects **174** from the second computing node **152a**. In this context, the first user **113** may store a first data file **132** at the first computing node **112a** owned and/or operated by the first user **113**. In the first data file **132**, the first user **113** may store the first type of data objects **134** owned by the first user **113**. Similarly, the second user **153** may store a second data file **172** at the second computing node **152a** owned and/or operated by the second user **153**. In the second data file **172**, the second user **153** may store the second type of data objects **174** owned by the second user **153**.

(47) To initiate the data exchange with the second computing node **152a**, the first computing node **112a** may be configured to transmit to the second computing node **152a**, a request **119** to perform a data interaction with the second computing node **152a**. The request **119** may indicate that the requested data interaction relates to exchanging data (e.g., data objects **134**, **174**) between the first user **113** and the second user **153**. In one embodiment, the first user **113** may trigger the generation and transmission of the request **119** by the first computing node **112a**.

(48) At operation **204**, if an acknowledgement **159** is not received from the second computing node **152a** (e.g., within a pre-configured time of transmitting the request **119**), method **200** proceeds to operation **206** where the first computing node **112a** discards the requested data interaction and the method **200** ends here. On the other hand, if an acknowledgement **159** is received from the second computing node **152a**, method **200** proceeds to operation **208**.

(49) As described above, in response to receiving the request **119** from the first computing node **112a**, the second computing node **152a** may generate and transmit an acknowledgement **159** to the first computing node **112a** indicating that the second computing node **152a** is ready to perform the requested data interaction with the first computing node **112a**. In one embodiment, the acknowledgement **159** may be automatically generated by the second computing node **152a** or manually triggered by the second user **153**. For example, the second user **153** may review the request **119** received by the second computing node **152a** and trigger the generation and transmission of the acknowledgement **159** to the first computing node **112a**.

(50) At operation **208**, first computing node **112a** generates a first smart contract **122**, an encrypted first network address **124**, and a first authentication code **126**.

(51) As described above, upon receiving the acknowledgement **159** from the second computing node **152a**, the first computing node **112a** may be configured to generate a first smart contract **122** that includes information relating to the data interaction. For example, the first smart contract **122** may include a first transformation rule **123** indicating that the first computing node **112a** is to transmit a first amount of the first type of data objects **134** belonging to the first user **113** in exchange for a second amount of the second type of data objects **174** belonging to the second user **153** that is to be received from the second computing node **152a**. In addition, the first computing node **112a** generates a first authentication code **126** that is unique to the data interaction initiated between the first computing node **112a** and the second computing node **152a**. The first authentication code **126** may include any known code such as an alphanumeric code. In addition, the first computing node **112a** generates an encrypted first network address **124** of a memory location within the first blockchain network **110** where the first type of data objects **134** associated with the first user **113** are stored. The first type of data objects **134** associated with the first user **113** may be stored at the first computing node **112a** or at any other computing node **112** of the first blockchain network **110**.

(52) At operation **210**, first computing node **112a** encrypts the first smart contract **122** using the first authentication code **126** to generate an encrypted first smart contract **122**.

(53) In one embodiment, the first computing node **112a** encrypts the first smart contract **122** based on an encrypt-decrypt algorithm previously agreed with the second computing node **152a**. As further described below, this allows the second computing node **152a** to decrypt the first smart contract **122** using the first authentication code **126** based on the same encrypt-decrypt algorithm previously agreed with the first computing node **112a**. Essentially, the first authentication code **126** acts as a unique key that can be used by the second computing node **152a** to access and review the first smart contract **122**.

(54) At operation **212**, first computing node **112a** transmits, to the second computing node **152a**, the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126**.

(55) In one embodiment, the first computing node **112a** transmits the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** in a single encoded first data file **120**. In alternative embodiments, the first computing node **112a** transmits the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** as separate data files.

(56) At operation **214**, if a first digital signature **168** is not received from the second computing node **152a** (e.g., within a pre-configured time of transmitting the first smart contract **122**), method **200** proceeds to operation **206** where the first computing node **112a** discards the requested data

interaction and the method **200** ends here. On the other hand, if a first digital signature **168** is received from the second computing node **152a**, method **200** proceeds to operation **216**.

(57) As described above, upon receiving the encrypted first smart contract **122**, the encrypted first network address **124**, and the first authentication code **126** from the first computing node **112a**, the second computing node **152a** may be configured to decrypt the first smart contract **122** using the first authentication code **126**. As described above, the second computing node **152a** may decrypt the first smart contract **122** based on the encrypt-decrypt algorithm previously agreed with the first computing node **112a**. After decrypting the first smart contract **122**, the second computing node **152a** may extract the first transformation rule **123** included in the first smart contract **122** and determine whether the first transformation rule **123** is acceptable. In one embodiment, the second computing node **152a** may determine whether the first transformation rule **123** is acceptable based on a pre-configured rule set for the second blockchain network **150** relating to data exchanges between a computing node **152** of the second blockchain network **150** and a computing node **112** of the first blockchain network. For example, the pre-configured rule set for the second blockchain network **150** may specify that at least X amount of the first type of data objects **134** is to be received for a unit amount of the second type of data objects **174**. The second computing node **152a** analyzes the first transformation rule **123** based on this pre-configured rule to determine whether the first transformation rule **123** is acceptable. As mentioned above, the first transformation rule **123** indicates that the first computing node **112a** is to transmit a first amount of the first type of data objects **134** belonging to the first user **113** in exchange for a second amount of the second type of data objects **174** belonging to the second user **153** that is to be received from the second computing node **152a**. When the first transformation rule **123** satisfies the pre-configured rule set for the second blockchain network **150**, the second computing node **152a** determines that the first transformation rule **123** is acceptable. For example, based on the first transformation rule **123**, when at least X amount of the first type of data objects **134** is to be received for a unit amount of the second type of data objects **174**, the second computing node **152a** determines that the first transformation rule **123** is acceptable and approves the first transformation rule **123**.

(58) It may be noted that the encrypted first network address **124** cannot be decrypted by the second computing node **152a** at this stage. The encrypted first network address **124** is configured to be decrypted by a unique access token (e.g., first access token **130**) which, as described below, the first computing node **112a** is configured to transmit to the second computing node **152a** at a later stage.

(59) In response to determining that the first transformation rule **123** is acceptable, the second computing node generates and transmits a first digital signature **168** to the first computing node **112a**. The first digital signature **168** indicates to the first computing node **112a** that the second computing node **152a** has approved the first transformation rule **123** that was included in the first smart contract **122**.

(60) At operation **216**, first computing node **112a** transmits a first access token **130** to the second computing node **152a**.

(61) As described above, after validating the first digital signature **168**, the first computing node transmits a first access token **130** to the second computing node **152a**. The first access token **130** is a key that can be used to decrypt the encrypted first network address **124** that was previously transmitted to the second computing node **152a**.

(62) At operation **218**, first computing node **112a** receives a request from the second computing node **152a** to transmit the first number of the first type of data objects **134** from the first memory location within the first blockchain network **110**.

(63) At operation **220**, first computing node **112a** initiates transmission of the first number of the first type of data objects **134** to the second computing node **152a**.

(64) As described above, the encrypted first network address **124** is of a memory location within

the first blockchain network **110** where the first type of data objects **134** associated with the first user **113** are stored. The second computing node **152a** receives the first access token **130** and decrypts the encrypted first network address **124** using the first access token **130** to determine the first network address **124**. Thereafter, the second computing node **152a** requests the first computing node **112a** to transmit the first amount of the first type of data objects **134** belonging to the first user **113** from the first network address **124**. For example, the second computing node **152a** transmits a request for the first amount of the first type of data objects **134** along with the first network address **124**. In response to receiving the request for the first type of data objects **134**, the first computing node **112a** initiates a transmission of the first amount of the first type of data objects **134** to the second computing node **152a**. In one embodiment, the second computing node **152a** transmits the request for the first type of data objects **134** to an exchange computing node (an exchange server in the second blockchain network **150**) in the second blockchain network **150** that is configured to coordinate data exchange between the first blockchain network **110** and the second blockchain network **150**. In an alternative or additional embodiment, the second computing node **152a** performs operations relating to the exchange computing node.

(65) The second computing node **152a** may be configured to transmit a second network address **164** of a memory location within the second blockchain network **150** where the second data objects **174** owned by the second user **153** are stored. In one embodiment, the second computing node **152a** transmits an encrypted second network address **164** and separately transmits a second access token **170** to the first computing node **112a**. The second access token **170** is a key that can be used to decrypt the encrypted second network address **164** that was previously transmitted to the first computing node **112a**. The first computing node **112a** receives the second access token **170** and decrypts the encrypted second network address **164** using the second access token **170** to determine the second network address **164**. Thereafter, the first computing node **112a** requests the second computing node **152a** to transmit the second amount of the second type of data objects **174** belonging to the second user **153** from the second network address **164**. For example, the first computing node **112a** transmits a request for the second amount of the second type of data objects **174** along with the second network address **164**. In response to receiving the request for the second type of data objects **174**, the second computing node **152a** initiates a transmission of the second amount of the second type of data objects **174** to the first computing node **112a**. In one embodiment, the first computing node **112a** transmits the request for the second type of data objects **174** to an exchange computing node (e.g., exchange server) within the first blockchain network **110** that is configured to coordinate data exchange between the first blockchain network **110** and the second blockchain network **150**. In an alternative or additional embodiment, the first computing node **112a** performs operations relating to the exchange computing node of the first blockchain network **110**. In one embodiment, the second computing node **152a** transmits the encrypted second network address **164** and the second access token **170** after approving the first transformation rule **123** included in the first smart contract **122**.

(66) In an example banking use case, the disclosed system and method may be used to exchange crypto currency between two different blockchain networks that support different type of crypto currency. For example, the first type of data objects may represent a first type of crypto currency and the second type of data objects may represent a second type of crypto currency. The first transformation rule **123** and the second transformation rule **163** may represent crypto exchange rates.

(67) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated in another system or certain features may be omitted, or not implemented.

(68) In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(69) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

Claims

1. A system comprising: a first computing node of a first blockchain network comprising a first plurality of computing nodes; and a processor communicatively coupled to the first computing node and configured to: transmit, to a second computing node of a second blockchain network comprising a second plurality of computing nodes, a request to perform a data interaction with the second computing node, wherein the data interaction comprises exchange of data between the first computing node and the second computing node; receive, from the second computing node, an acknowledgement indicating that the second computing node is ready to perform the data interaction with the first computing node; in response to receiving the acknowledgement: generate a first smart contract comprising information relating to the data interaction, wherein the information relating to the data interaction at least comprises a first transformation rule indicating that a first number of a first type of data objects is to be transmitted by the first computing node in exchange for a second number of a second type of data objects received from the second computing node; generate an encrypted first network address of a first memory location within the first blockchain network where the first type of data objects is stored; and generate a first authentication code for the data interaction; encrypt the first smart contract using the first authentication code to generate an encrypted first smart contract; transmit to the second computing node the encrypted first smart contract, the encrypted first network address, and the first authentication code, wherein the second computing node decrypts the first smart contract using the first authentication code; receive a first digital signature from the second computing node, wherein the first digital signature indicates that the second computing node has approved the first transformation rule included in the first smart contract; after receiving the first digital signature, transmit to the second computing node a first access token that is to be used to decrypt the encrypted first network address of the first memory location within the first blockchain network where the first type of data objects is stored; receive, from the second computing node, a first request for transmission of the first number of the first type of data objects from the first memory location within the first blockchain network; and in response to receiving the request, initiate transmission of the first number of the first type of data objects to the second computing node from the first memory location within the first blockchain network.

2. The system of claim 1, wherein the processor is further configured to: encode the encrypted first smart contract, the encrypted first network address, and the first authentication code using a synthetic DNA synthesis process to generate an encoded first data file, wherein the encoded first data file comprises one or more synthetic DNA sequences that represent encoded data; and transmit the encoded first data file to the second computing node, wherein the second computing node decodes the one or more synthetic DNA sequences in the encoded first data file to extract the encrypted first smart contract, the encrypted first network address, and the first authentication code.

3. The system of claim 1, wherein the processor is further configured to: receive from the second computing node: a second smart contract comprising information relating to the data interaction, wherein the information comprised in the second smart contract at least comprises a second transformation rule indicating the second number of the second type of data objects to be transmitted by the second computing node to the first computing node in exchange for the first number of the first type of data objects to be received by the second computing node from the first computing node; an encrypted second network address of a second memory location within the second blockchain network where the second type of the data objects is stored; and a second authentication code for the data interaction; decrypt the second smart contract using the second authentication code; receive an indication that the second transformation rule in the second smart contract is acceptable; in response to receiving the indication, generate a second digital signature; transmit the second digital signature and the second authentication code to the second computing node, wherein the second digital signature indicates that the first computing node has approved the second transformation rule included the second smart contract; receive, from the second computing node, a second access token that is to be used to decrypt the encrypted second network address of the second memory location within the second blockchain network where the second type of the data objects is stored; decrypt the encrypted second network address using the second access token; transmit, to the second computing node, a second request for transmission of the second number of the second type of the data objects from the second network address in the second blockchain network; and after transmitting the second request, receive, from the second computing node, the second number of the second type of data objects from the second memory location within the second blockchain network.

4. The system of claim 3, wherein the processor is further configured to: receive an encoded second data file from the second computing node, wherein the encoded second data file is encoded using a synthetic DNA synthesis process and comprises one or more synthetic DNA sequences representing the encrypted second smart contract, the encrypted second network address, and the second authentication code; and decode the one or more synthetic sequences in the encoded second data file to extract the encrypted second smart contract, the encrypted second network address, and the second authentication code.

5. The system of claim 1, wherein the processor is further configured to: receive, from the second computing node, the first digital signature along with the first authentication code that was previously transmitted to the second computing node; and validate the first digital signature by matching the first digital signature received from the second computing node with the first digital signature transmitted to the second computing node.

6. The system of claim 5, wherein the processor is further configured to: transmit the first access token to the second computing node in response to successfully validating the first digital signature.

7. The system of claim 5, wherein the first digital signature and the first authentication code is received from the second computing node with the first smart contract that was previously transmitted to the second computing node.

8. A method for performing a data interaction, the method comprising: transmit, to a second computing node of a second blockchain network comprising a second plurality of computing nodes, a request to perform a data interaction with the second computing node, wherein the data interaction comprises exchange of data between a first computing node of a first blockchain network and the second computing node, wherein the first blockchain network comprises a first plurality of computing nodes; receive, from the second computing node, an acknowledgement indicating that the second computing node is ready to perform the data interaction with the first computing node; in response to receiving the acknowledgement: generate a first smart contract comprising information relating to the data interaction, wherein the information relating to the data interaction at least comprises a first transformation rule indicating that a first number of a first type

of data objects is to be transmitted by the first computing node in exchange for a second number of a second type of data objects received from the second computing node; generate an encrypted first network address of a first memory location within the first blockchain network where the first type of data objects is stored; and generate a first authentication code for the data interaction; encrypt the first smart contract using the first authentication code to generate an encrypted first smart contract; transmit to the second computing node the encrypted first smart contract, the encrypted first network address, and the first authentication code; receive a first digital signature from the second computing node, wherein the first digital signature indicates that the second computing node has approved the first transformation rule included in the first smart contract; after receiving the first digital signature, transmit to the second computing node a first access token that is to be used to decrypt the encrypted first network address of the first memory location within the first blockchain network where the first type of data objects is stored; receive, from the second computing node, a first request for transmission of the first number of the first type of data objects from the first memory location within the first blockchain network; and in response to receiving the request, initiate transmission of the first number of the first type of data objects to the second computing node from the first memory location within the first blockchain network.

9. The method of claim 8, further comprising: encoding the encrypted first smart contract, the encrypted first network address, and the first authentication code using a synthetic DNA synthesis process to generate an encoded first data file, wherein the encoded first data file comprises one or more synthetic DNA sequences that represent encoded data; and transmitting the encoded first data file to the second computing node, wherein the second computing node decodes the one or more synthetic DNA sequences in the encoded first data file to extract the encrypted first smart contract, the encrypted first network address, and the first authentication code.

10. The method of claim 8, further comprising: receiving from the second computing node: a second smart contract comprising information relating to the data interaction, wherein the information comprised in the second smart contract at least comprises a second transformation rule indicating the second number of the second type of data objects to be transmitted by the second computing node to the first computing node in exchange for the first number of the first type of data objects to be received by the second computing node from the first computing node; an encrypted second network address of a second memory location within the second blockchain network where the second type of the data objects is stored; and a second authentication code for the data interaction; decrypt the second smart contract using the second authentication code; receive an indication that the second transformation rule in the second smart contract is acceptable; in response to receiving the indication, generating a second digital signature; transmitting the second digital signature and the second authentication code to the second computing node, wherein the second digital signature indicates that the first computing node has approved the second transformation rule included in the second smart contract; receiving, from the second computing node, a second access token that is to be used to decrypt the encrypted second network address of the second memory location within the second blockchain network where the second type of the data objects is stored; decrypting the encrypted second network address using the second access token; transmitting, to the second computing node, a second request for transmission of the second number of the second type of the data objects from the second network address in the second blockchain network; and after transmitting the second request, receiving, from the second computing node, the second number of the second type of data objects from the second memory location within the second blockchain network.

11. The method of claim 10, further comprising: receiving an encoded second data file from the second computing node, wherein the encoded second data file is encoded using a synthetic DNA synthesis process and comprises one or more synthetic DNA sequences representing the encrypted second smart contract, the encrypted second network address, and the second authentication code; and decoding the one or more synthetic sequences in the encoded second data file to extract the

encrypted second smart contract, the encrypted second network address, and the second authentication code.

12. The method of claim 8, further comprising: receiving, from the second computing node, the first digital signature along with the first authentication code that was previously transmitted to the second computing node; and validating the first digital signature by matching the first digital signature received from the second computing node with the first digital signature transmitted to the second computing node.

13. The method of claim 12, further comprising: transmitting the first access token to the second computing node in response to successfully validating the first digital signature.

14. The method of claim 12, wherein the first digital signature and the first authentication code is received from the second computing node with the first smart contract that was previously transmitted to the second computing node.

15. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: transmit, to a second computing node of a second blockchain network comprising a second plurality of computing nodes, a request to perform a data interaction with the second computing node, wherein the data interaction comprises exchange of data between a first computing node of a first blockchain network and the second computing node, wherein the first blockchain network comprises a first plurality of computing nodes; receive, from the second computing node, an acknowledgement indicating that the second computing node is ready to perform the data interaction with the first computing node; in response to receiving the acknowledgement: generate a first smart contract comprising information relating to the data interaction, wherein the information relating to the data interaction at least comprises a first transformation rule indicating that a first number of a first type of data objects is to be transmitted by the first computing node in exchange for a second number of a second type of data objects received from the second computing node; generate an encrypted first network address of a first memory location within the first blockchain network where the first type of data objects is stored; and generate a first authentication code for the data interaction; encrypt the first smart contract using the first authentication code to generate an encrypted first smart contract; transmit to the second computing node the encrypted first smart contract, the encrypted first network address, and the first authentication code; receive a first digital signature from the second computing node, wherein the first digital signature indicates that the second computing node has approved the first transformation rule included in the first smart contract; after receiving the first digital signature, transmit to the second computing node a first access token that is to be used to decrypt the encrypted first network address of the first memory location within the first blockchain network where the first type of data objects is stored; receive, from the second computing node, a first request for transmission of the first number of the first type of data objects from the first memory location within the first blockchain network; and in response to receiving the request, initiate transmission of the first number of the first type of data objects to the second computing node from the first memory location within the first blockchain network.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to: encode the encrypted first smart contract, the encrypted first network address, and the first authentication code using a synthetic DNA synthesis process to generate an encoded first data file, wherein the encoded first data file comprises one or more synthetic DNA sequences that represent encoded data; and transmit the encoded first data file to the second computing node, wherein the second computing node decodes the one or more synthetic DNA sequences in the encoded first data file to extract the encrypted first smart contract, the encrypted first network address, and the first authentication code.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to: receive from the second computing node: a second smart contract comprising information relating to the data interaction, wherein the information comprised in the

second smart contract at least comprises a second transformation rule indicating the second number of the second type of data objects to be transmitted by the second computing node to the first computing node in exchange for the first number of the first type of data objects to be received by the second computing node from the first computing node; an encrypted second network address of a second memory location within the second blockchain network where the second type of the data objects is stored; and a second authentication code for the data interaction; decrypt the second smart contract using the second authentication code; receive an indication that the second transformation rule in the second smart contract is acceptable; in response to receiving the indication, generate a second digital signature; transmit the second digital signature and the second authentication code to the second computing node, wherein the second digital signature indicates that the first computing node has approved the second transformation rule included the second smart contract; receive, from the second computing node, a second access token that is to be used to decrypt the encrypted second network address of the second memory location within the second blockchain network where the second type of the data objects is stored; decrypt the encrypted second network address using the second access token; transmit, to the second computing node, a second request for transmission of the second number of the second type of the data objects from the second network address in the second blockchain network; and after transmitting the second request, receive, from the second computing node, the second number of the second type of data objects from the second memory location within the second blockchain network.

18. The non-transitory computer-readable medium of claim 17, wherein the instructions further cause the processor to: receive an encoded second data file from the second computing node, wherein the encoded second data file is encoded using a synthetic DNA synthesis process and comprises one or more synthetic DNA sequences representing the encrypted second smart contract, the encrypted second network address, and the second authentication code; and decode the one or more synthetic sequences in the encoded second data file to extract the encrypted second smart contract, the encrypted second network address, and the second authentication code.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions further cause the processor to: receive, from the second computing node, the first digital signature along with the first authentication code that was previously transmitted to the second computing node; and validate the first digital signature by matching the first digital signature received from the second computing node with the first digital signature transmitted to the second computing node.

20. The non-transitory computer-readable medium of claim 19, wherein the instructions further cause the processor to: transmit the first access token to the second computing node in response to successfully validating the first digital signature.
